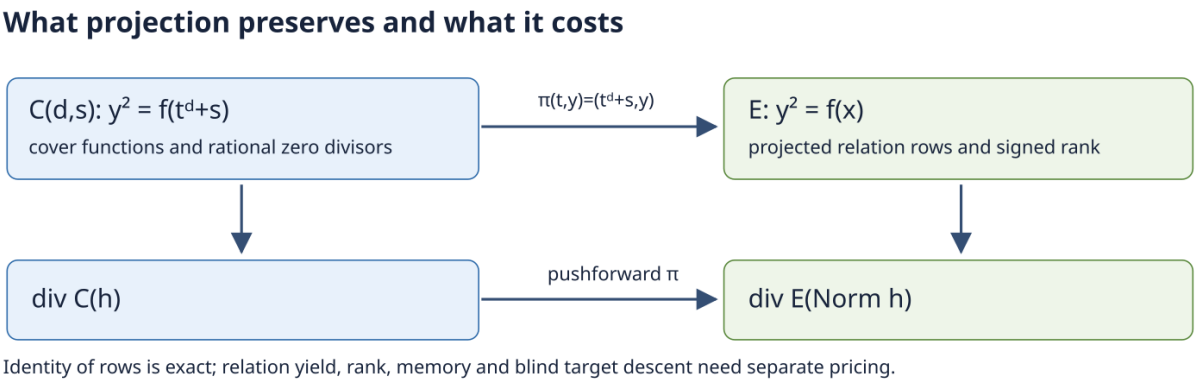
Projected Relations from Prime-Field Hyperelliptic Covers

Reproducible toy study

24 October 2026

Question. Can the covers $C_{d,s} : y^2 = f(t^d + s)$ of an elliptic curve $E : y^2 = f(x)$ supply reusable projected relations and blind-target descent with total work below $\sqrt{\ell}$? This report tests one bounded linear-section family on a public $\mathbb{F}_{101}$ fixture. It does not implement Enge's high-genus Jacobian algorithm or establish an asymptotic bound.



Algebraic reduction

For $p > 3$, d prime to p, and $f(s) \neq 0$, the cover is smooth and has genus $(3d-1)/2$ for odd d or $(3d-2)/2$ for even d. The degree-d map $\pi(t, y) = (t^d + s, y)$ gives $\pi_*\pi^* = [d]$ on $E = \mathrm{Jac}(E)$. The divisor norm identity is $\pi_*(\mathrm{div}_C\, h) = \mathrm{div}_E\, N(h)$. For $h = y - (at + b)$,

$$N(h) = (y - b)^d - a^d(x - s).$$

For the degree-two target section $h = y - y_Q - (t^2 + s - x_Q)(ct + b)$, with $A = y - y_Q - b(x - x_Q)$, the direct elliptic norm is

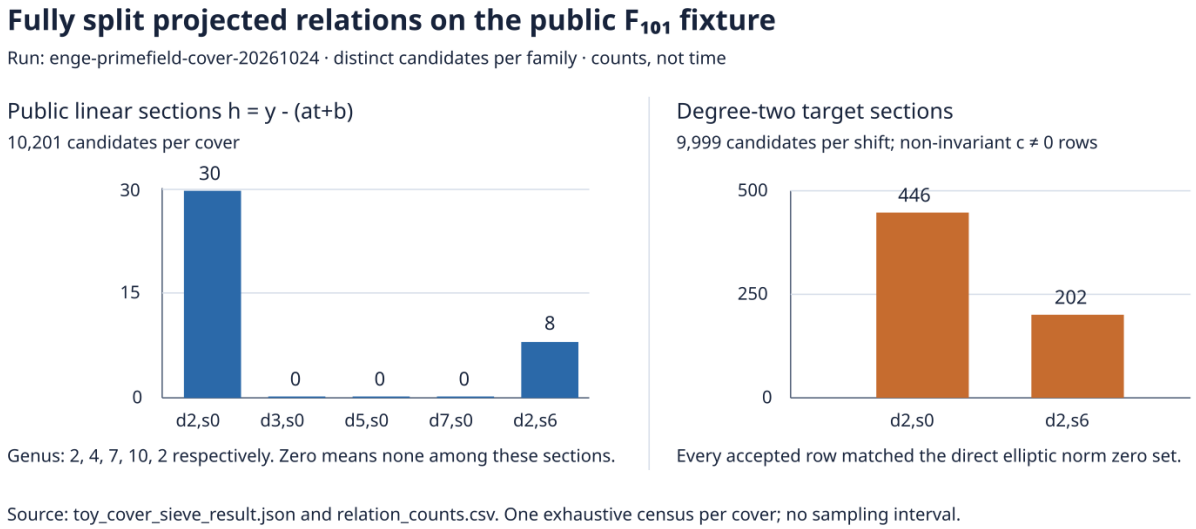
$$N(h) = A^2 - c^2(x - s)(x - x_Q)^2.$$

The degree-two product identity has an evaluated `identity.certificate/v1` record and Lean proof source in the study directory. The Lean build was not verified in this run; the source does not formalize the geometric norm map.

Exhaustive public fixture

The curve is $E/\mathbb{F}_{101} : y^2 = x^3 + x + 46$, with $\#E = 109$, $G = (2, 37)$, and $Q = (64, 19)$. The Rust collector does not take the target scalar as input. Each cover scans all 101^2 sections $y - (at + b)$, retaining a row only when its full finite zero divisor splits into distinct rational points. Target sections are scanned only when the target fiber is fully rational. The table gives counts, not timings.

(d, s)	genus	projected points	full public splits	signed rank/columns
$(2, 0)$	2	56	30	9/25
$(3, 0)$	4	108	0	0/2
$(5, 0)$	7	20	0	0/2
$(7, 0)$	10	108	0	0/2
$(2, 6)$	2	44	8	2/11



For $(d, s) = (2, 6)$, target-only streaming recovered $\log_G Q = 37$ after 1,070 candidate sections, 108,070 cover-point tests, and 22 accepted rows. The signed relation matrix had rank 21 on 22 columns; anchoring $\log G = 1$ determined $\log Q$. A separate multiplication verified $37G = Q$. Across the full target census, all 202 non-invariant fully split rows matched the direct elliptic norm zeros, and the same 202 parameter pairs were accepted directly on E.

The deterministic rho control reached a collision after 72 elliptic group operations. Cover-point tests, rank solves, and group operations are unlike units, so these figures are not a calibrated speed ratio. The shift $s = 6$ was chosen using public G, Q; that selection cost was not measured. No below-rho conclusion follows. At genera 4, 7, and 10, zero relations were found only within this fully rational linear-section factor base; higher-degree primes and Enge's full algorithm are untested.

Replay and status

The native run record, root result copy, and exact replay commands are listed in the adjacent `REPORT.md`. Run the study crate's release tests and release binary with Cargo. The collector verifies each accepted row by elliptic addition, each direct norm zero set, and the recovered scalar by multiplication. The open next step is a degree-$B > 1$ cover-prime factor base with arbitrary-divisor norm, projected rank, target descent, and one-target same-host cost and memory accounting.